SIMATS ENGINEERING
ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME: COMPUTER VISION **COURSE CODE: ITA05**

COURSE OUTCOME COVERED

CO4: *Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics.* (BTL 5)

Assessment Tool Weightage:20%

Analyze the given problem by comparing available techniques against the specified constraints and requirements. Evaluate and justify your decision using logical reasoning, clearly indicating how the chosen approach satisfies the conditions.

1. Smart Surveillance System Selection
A city surveillance system must operate under the following conditions:

- If latency > 100 ms → system fails
- If accuracy < 85% → unacceptable
- Hardware supports only moderate computation

Models available:

Viola-Jones: 78%, low latency
YOLO: 92%, moderate latency
Deep Learning: 95%, high latency

Compare the models and evaluate which satisfies all constraints. Justify your decision using logical reasoning based on given conditions.

2. Autonomous Vehicle Detection Logic
A vehicle detection system must:

- Prioritize safety (accuracy > 90%)
- Maintain response time < 50 ms

Options:

Method A: 88% accuracy, 30 ms
Method B: 93% accuracy, 70 ms

Method C: 91% accuracy, 45 ms

Compare the methods and evaluate which should be selected. Justify your choice using constraint-based reasoning.

3. Motion Analysis Decision Problem

A traffic system requires:

- Accurate motion direction detection
- Robustness in dense traffic

Results:

Optical Flow: accurate but noisy in dense scenes

Motion Estimation: stable but less detailed

Compare both approaches and evaluate which is suitable under high-density conditions. Justify using scenario-based logic.

4. Background Modeling Evaluation

A GMM-based system produces:

- Static scenes: 90% accuracy
- Dynamic scenes: 65% accuracy

Requirement:

System must maintain ≥80% accuracy across all scenarios

Compare performance and evaluate whether GMM alone meets requirements. Justify logically and suggest a decision.

5. Depth Estimation Strategy

Constraints:

- Only one camera available
- Moderate depth accuracy acceptable

Options:

Stereo Vision: high accuracy, needs two cameras

Structure from Motion: moderate accuracy, single camera

Compare both approaches and evaluate which is feasible. Justify using constraint-based reasoning.

6. Retail Motion Tracking Logic

A retail system requires:

- Accurate tracking when crowd density is low
- Stable tracking when density is high

Observations:

Optical Flow: accurate in low density, noisy in high density

Motion Models: stable in high density, less detailed

Compare both and evaluate a suitable approach or combination. Justify your reasoning logically.

7. Defect Detection System Decision
Requirements:
- Detect subtle defects (>95% accuracy)
- Adapt to product variations

Options:

Traditional methods: fast, less adaptive
Deep learning: accurate, requires training

Compare both approaches and evaluate which meets requirements. Justify using performance vs adaptability logic.

8. AR Motion Tracking Constraint Problem
An AR system must:
- Maintain stable tracking (<5% drift)
- Run on mobile hardware

Options:

Optical Flow: detailed but computationally intensive
Motion Estimation: less detailed but efficient

Compare and evaluate which method satisfies both constraints. Justify your decision logically.

9. Mobile Face Detection Trade-off
Constraints:
- Real-time operation required
- Accuracy ≥85%

Options:

Viola-Jones: 80%, fast
Deep Learning: 95%, slow

Compare and evaluate which approach should be chosen. Justify using constraint satisfaction logic.

10. Animation System Optimization
A game engine requires:
- Realistic motion for cutscenes
- Fast rendering for gameplay

Options:

Motion Models: realistic, computationally heavy
Motion Estimation: efficient, less realistic

Compare both approaches and evaluate how the system should be designed. Justify using scenario-based logical reasoning.

11. Traffic Surveillance Model Selection

A traffic monitoring system must:

- Accuracy ≥ 90%
- Latency ≤ 80 ms
- Works under varying lighting

Options:
Method A: 88%, 60 ms
Method B: 91%, 85 ms
Method C: 92%, 75 ms

Compare and evaluate which method satisfies all constraints. Justify logically.

12. Drone Vision System Constraint

A drone system requires:

- Lightweight computation
- Real-time detection (<60 ms)
- Accuracy ≥ 85%

Options:
Model A: 83%, 40 ms
Model B: 87%, 65 ms
Model C: 86%, 55 ms

Compare and evaluate the suitable model. Justify using constraints.

13. Security System Recognition

A security system must:

- High accuracy (>92%)
- Moderate computation

Options:
Traditional: 85%, low computation
Deep Learning: 94%, high computation
Hybrid: 91%, moderate computation

Compare and evaluate the best approach. Justify logically.

14. Motion Tracking in Sports Analytics

Requirements:

- Accurate motion tracking
- Real-time performance

Options:
Optical Flow: accurate, slow
Motion Estimation: fast, less accurate

Compare and evaluate the suitable approach. Justify using constraints.

15. Face Recognition System Design

Constraints:

- Accuracy ≥ 90%
- Works in low light
- Moderate computation

Options:
Traditional: 82%
Deep Learning: 93%
Hybrid: 89%

Compare and evaluate the best choice. Justify logically.

16. Autonomous Navigation Constraint

System must:

- Accuracy ≥ 95%
- Latency < 70 ms

Options:
Model A: 94%, 60 ms
Model B: 96%, 80 ms
Model C: 95%, 65 ms

Compare and evaluate. Justify your decision.

17. Industrial Inspection System

Requirements:

- Detect defects with ≥ 95% accuracy
- Real-time operation

Options:

Method A: 92%, fast
Method B: 96%, slow
Method C: 95%, moderate

Compare and evaluate the suitable method. Justify your decision.

18. Crowd Monitoring System

Requirements:

- Stable detection in dense crowds
- Moderate accuracy acceptable

Options:
Optical Flow: unstable in dense scenes
Motion Models: stable

Compare and evaluate. Justify your decision.

19. Video Analytics Pipeline

Constraints:

- Accuracy ≥ 90%
- Processing time < 100 ms

Options:
Pipeline A: 88%, 80 ms
Pipeline B: 91%, 120 ms
Pipeline C: 90%, 95 ms

Compare and evaluate the best pipeline. Justify your decision.

20. Smart Parking System

Requirements:

- Detect vehicles accurately
- Low computational cost

Options:
Viola-Jones: low cost, low accuracy
YOLO: moderate cost, high accuracy

Compare and evaluate. Justify your decision.

21. Medical Imaging Analysis

Requirements:

- Very high accuracy (>95%)
- Processing time not critical

Options:
Traditional: 85%
Deep Learning: 97%

Compare and evaluate the suitable approach. Justify your decision.

22. Edge Device Vision System

Constraints:

- Low power
- Real-time performance

Options:
Deep Learning: accurate, heavy
Lightweight Model: moderate accuracy, efficient

Compare and evaluate. Justify logically.

23. Traffic Flow Analysis

Requirements:

- Detect motion patterns
- Robust in crowded scenes

Options:
Optical Flow: detailed, noisy
Motion Estimation: stable

Compare and evaluate. Justify your decision.

24. Retail Analytics System

Requirements:

- Customer tracking

- Moderate accuracy
- Real-time processing

Options:
Method A: accurate, slow
Method B: moderate accuracy, fast

Compare and evaluate. Justify your decision.

25. Object Detection in Foggy Conditions

Requirements:

- Robust under poor visibility
- Accuracy ≥ 85%

Options:
Model A: 80%
Model B: 87%
Model C: 90% but slow

Compare and evaluate. Justify your decision.

26. Video Surveillance Upgrade

Constraints:

- Improve accuracy
- Maintain same latency

Options:
Model A: 85%, 50 ms
Model B: 90%, 50 ms

Compare and evaluate. Justify your decision.

27. Robotics Vision System

Requirements:

- Fast decision-making
- Moderate accuracy

Options:
Deep Learning: slow
Traditional: fast

Compare and evaluate. Justify your decision.

28. Smart Home Security

Requirements:

- Detect intrusions accurately
- Low computation

Options:
YOLO: accurate
Viola-Jones: efficient

Compare and evaluate. Justify your decision.

29. Autonomous Drone Tracking

Constraints:

- Real-time (<40 ms)
- Accuracy ≥ 85%

Options:
Model A: 83%, 30 ms
Model B: 86%, 50 ms
Model C: 88%, 38 ms

Compare and evaluate. Justify your decision.

30. Motion Animation System

Requirements:

- Smooth motion
- Real-time rendering

Options:
Motion Models: realistic, slow
Motion Estimation: fast, less realistic

Compare and evaluate design approach. Justify your decision.

Code of Conduct:

I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weigh tage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough	Good	Basic	Poor or no

		analysis with validated results and performance evaluation	analysis with reasonable validation	analysis with limited validation	analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

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5. Depth Estimation Strategy

Given Constraints

- Only one camera is available.
- Moderate depth accuracy is acceptable.

Comparison

Feature	Stereo Vision	Structure from Motion
Cameras required	Two	One
Depth accuracy	High	Moderate
Computational requirement	Higher	Moderate
Single-camera feasibility	No	Yes
Suitable for given system	No	Yes

Evaluation

Stereo Vision calculates depth using the difference between images captured from two cameras. It can provide high accuracy, but it violates the constraint that only one camera is available.

Structure from Motion (SfM) estimates depth by analyzing the movement of a single camera across multiple frames. It does not require two cameras and provides moderate depth accuracy, which is sufficient for the given requirement.

Conclusion

Structure from Motion is the feasible choice.

The decision is based directly on the constraints:

- Only one camera → SfM satisfies this requirement.
- Moderate accuracy acceptable → SfM provides sufficient accuracy.
- Stereo Vision cannot be selected because it requires two cameras.

Therefore, SfM provides the best constraint-compatible solution.

6. Retail Motion Tracking Logic

Requirements

- Accurate tracking when crowd density is low.
- Stable tracking when crowd density is high.

Given Observations

Method	Low Crowd Density	High Crowd Density
Optical Flow	Accurate	Noisy
Motion Models	Less detailed	Stable

Evaluation

Optical Flow

- Tracks motion by analyzing changes in pixel intensity between consecutive frames.
- Works accurately when people are separated.
- In crowded environments, overlapping objects and similar movements can make tracking noisy.

Motion Models

- Predict object movement using a model of motion.
- Less detailed than optical flow.
- More stable when many objects are present.
- Useful for maintaining tracking when objects overlap or temporarily disappear.

Best Approach: Combination

A hybrid approach is most suitable.

- When crowd density is low, use Optical Flow for accurate and detailed tracking.
- When crowd density is high, use Motion Models for stable tracking.
- The system can switch between the two methods based on detected crowd density.

Conclusion

A combination of both techniques provides the best solution:

Low Density → Optical Flow

High Density → Motion Models

This approach combines the accuracy of optical flow with the stability of motion models, making it suitable for a retail environment with changing crowd conditions.

11. Traffic Surveillance Model Selection

Requirements

- Accuracy ≥ 90%
- Latency ≤ 80 ms
- Works under varying lighting

Given Results

Method	Accuracy	Latency	Meets Accuracy?	Meets Latency?
A	88%	60 ms	No	Yes
B	91%	85 ms	Yes	No
C	92%	75 ms	Yes	Yes

Evaluation

Method A

- Accuracy = 88%, which is below the required 90%.
- Latency = 60 ms, which satisfies the latency requirement.
- Therefore, it fails the accuracy constraint.

Method B

- Accuracy = 91%, satisfying the accuracy requirement.
- Latency = 85 ms, exceeding the maximum allowed 80 ms.
- Therefore, it fails the latency constraint.

Method C

- Accuracy = 92% ≥ 90%
- Latency = 75 ms ≤ 80 ms
- It satisfies both quantitative constraints.
- Assuming its performance under varying lighting is acceptable as specified by the system requirement, it is the only suitable option.

Conclusion

Method C is the best choice.

It satisfies all the stated numerical constraints: 92% ≥ 90% and 75 ms ≤ 80 ms.

Therefore, Method C should be selected for the traffic surveillance system. Method A fails accuracy, while Method B fails latency.

28. Smart Home Security

Requirements

- Detect intrusions accurately.
- Use low computational resources.

Comparison

Feature	YOLO	Viola-Jones
Detection capability	High	Moderate
Accuracy	High	Lower for general objects
Computational efficiency	Moderate/High resource requirement	Efficient
General object detection	Excellent	Limited
Low-computation suitability	Lower	High

Evaluation

YOLO

- YOLO is a modern deep-learning-based object detection method.
- Provides high detection accuracy.
- Can detect different types of objects and people.
- However, it generally requires more computational resources.

Viola-Jones

- Uses Haar-like features and a cascade classifier.
- Computationally efficient.
- Suitable for systems with limited processing power.
- However, its detection capability is more limited compared with modern deep-learning detectors.

Decision

Since the system requires both accurate intrusion detection and low computation, there is a trade-off.

If accuracy is the dominant requirement, YOLO is preferable.

If the hardware is severely computation-limited, Viola-Jones is more practical.

For a smart security system where missing an intruder is costly, a lightweight YOLO model would be a strong practical choice if the hardware can support it.

Conclusion

YOLO is preferred when accurate intrusion detection is the priority, while Viola-Jones is preferred when computational efficiency is the main constraint. For a security-critical system, accuracy should generally receive higher priority, making a lightweight YOLO-based solution preferable.

30. Motion Animation System

Requirements

- Smooth motion.
- Real-time rendering.

Given Options

Feature	Motion Models	Motion Estimation
Realism	High	Lower
Speed	Slow	Fast
Smooth real-time rendering	Limited	Suitable
Computational cost	High	Lower
Real-time suitability	Low	High

Motion Models

Motion models generate realistic movement based on physical or mathematical models.

Advantages:

- Highly realistic motion.
- Natural-looking movement.
- Good physical consistency.

Disadvantage:

- Computationally expensive.
- Slow for real-time rendering.

Motion Estimation

Motion estimation determines movement between frames efficiently.

Advantages:

- Fast processing.
- Suitable for real-time systems.
- Can maintain high frame rates.
- Lower computational requirements.

Disadvantage:

- Less physically realistic.
- May produce simplified motion.

Design Decision

Since the system specifically requires real-time rendering, speed is a critical constraint.

Therefore: Motion Estimation → Primary Technique

A practical design can also use a hybrid approach, where motion estimation provides fast real-time movement and simplified motion models are applied where additional realism is required.

Conclusion

Motion Estimation is the better primary choice because it satisfies the real-time requirement. Although Motion Models provide greater realism, their slow processing makes them less suitable for real-time animation. A hybrid system can provide a balance between smoothness, speed, and realism.